e-psu rev A

USB-controlled precision power stage for a bench PSU: 0-24 V / 0-3 A CV/CC with uA-resolution current ranging,
plus a measured 10 A pass-through. github.com/embedded-pro/e-psu

Power input & housekeeping rails

30 V input, reverse-polarity P-FET, TVS, fuses
LMR16006X 30 V -> 5 V housekeeping buck
+3V3 (AMS1117), +3.3VA (TLV755), +5VA
VOP filter, optional LM2776 VNEG

File: power_input.kicad_sch

Tracking buck pre-regulator

LM76003 synchronous buck, 500 kHz
VPRE tracks VOUT + 1.5 V via DAC injection
into FB (BUCK_DAC from PA4)
Optional V_GATE charge pump (not fitted)

File: buck.kicad_sch

Linear CV/CC stage & setpoint DAC

IRFP260N source follower + 15 mOhm CC shunt
OPA2192 CV / CC error amps, OPA192 buffer
DAC80502 16-bit setpoints (SPI)
Gate hold / OCP clamp, no-load bleeder

File: linear.kicad_sch

Current measurement & auto-ranging

1 R + 100 R ranging shunts with BAV199 clamps
AO3400A bypass FETs on a VFORCE island
(B05055, CA-IS3720HS, UCC27524A)
INA190A3 range amplifiers

File: ranging.kicad_sch

Pass-through & output switch

10 A pass-through: 2 mOhm + INA281A3
HF115F mode relay (NC = precision path)
Back-to-back NTMFS5C430N output switch
VOM1271 driver, output TVS, VOUT_TERM

File: output.kicad_sch

MCU, debug & user interface

STM32G474RET6, 8 MHz crystal, SWD
COMP + TIM break for us range-up / OCP
Buttons, 4 status LEDs, board NTC

File: mcu.kicad_sch

Isolated USB

USB-C (USB 2.0 FS), USBLC6 ESD
ADuM3160 2.5 kV isolator
PC side powered from VBUS

File: usb.kicad_sch

Power flow: bench PSU (30 V) -> input protection -> tracking buck (VPRE = VOUT + 1.5 V) -> linear CV/CC stage
-> force node -> ranging shunts (1 R / 100 R) -> sense node -> mode relay -> back-to-back output switch -> OUT
Pass-through: bench PSU -> 15 A fuse -> 2 mOhm shunt -> mode relay (NO) -> output switch -> OUT

Ground domains: GND (main board) | GND_USB (PC side of U702) | VFORCE island (range-drive, floats with the output).
Generated schematic: run ERC in KiCad 7+ after opening. U201 footprint (TI RNP0030B) must be imported.

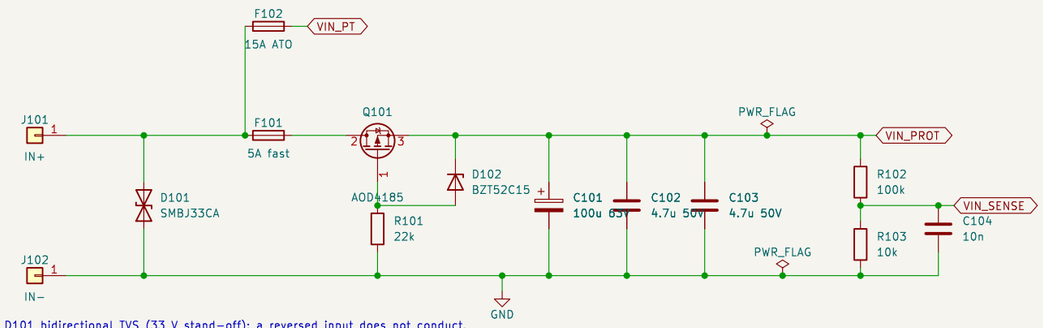
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Sheet: /
File: e-psu.kicad_sch

Title: e-psu: precision USB power stage

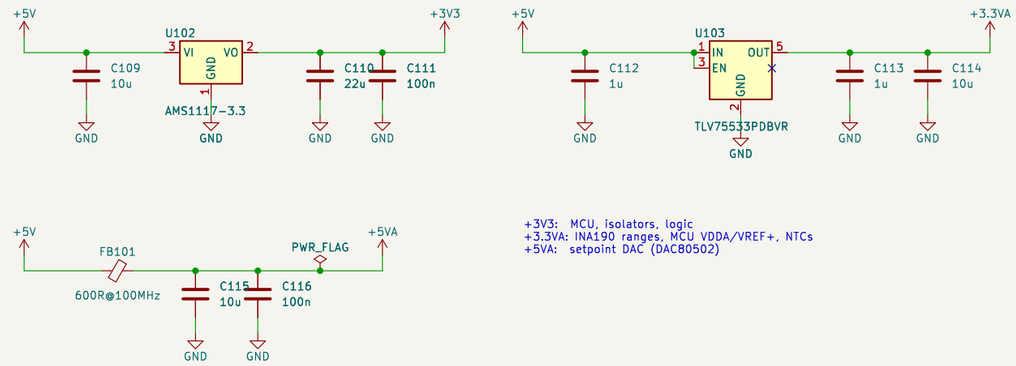
Size: A3	Date: 2026-11-03	**Rev: A1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/8

Input & protection (30 V from the bench PSU)



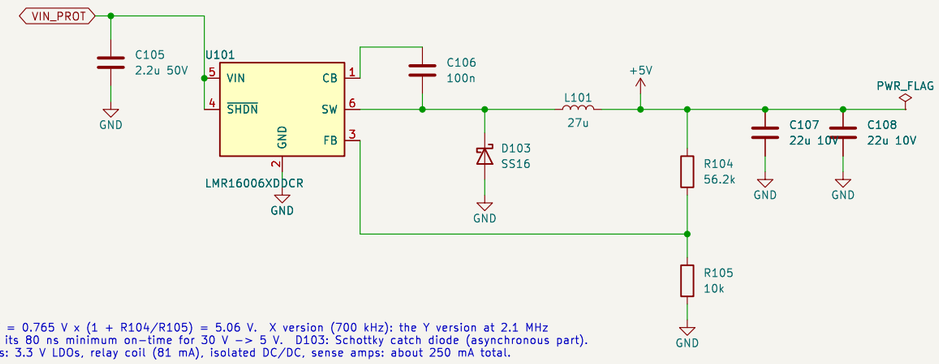
D101 bidirectional TVS (33 V stand-off): a reversed input does not conduct.
Q101 blocks a reversed input on the precision path; D102 limits VGS to −15 V.
F102 / VIN_PT feed the 10 A pass-through shunt (output sheet) directly.
VIN_SENSE = VIN x 10k / 110k (30 V -> 2.73 V)

Logic and analog rails



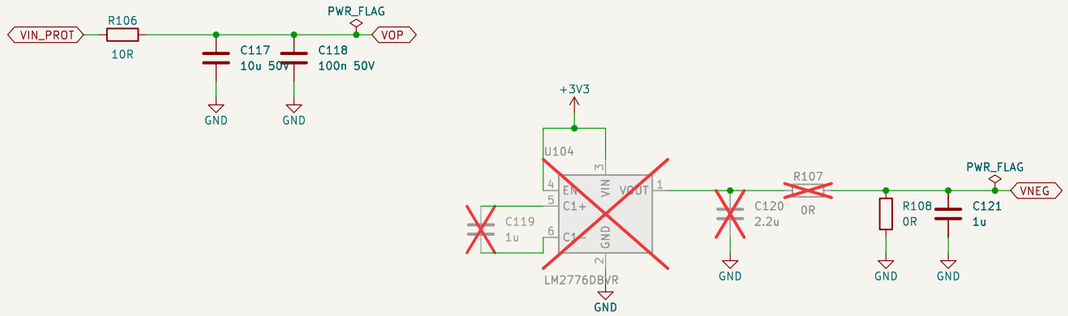
+3V3: MCU, isolators, logic
+3.3VA: INA190 ranges, MCU VDDA/VREF+, NTCs
+5VA: setpoint DAC (DAC80502)

Housekeeping buck 30 V -> 5 V (0.6 A)



VOUT = 0.765 V x (1 + R104/R105) = 5.06 V. X version (700 kHz): the Y version at 2.1 MHz
is at its 80 ns minimum on-time for 30 V -> 5 V. D103: Schottky catch diode (asynchronous part).
Loads: 3.3 V LDOs, relay coil (81 mA), isolated DC/DC, sense amps: about 250 mA total.

Op-amp supplies (VOP / VNEG)



Default (single supply): R108 fitted, U104 / C119 / C120 / R107 not fitted -> VNEG = GND.
For a true 0 V output: fit U104, C119, C120, R107 and remove R108 (VNEG = −3.2 V).
VOP + |VNEG| must stay <= 36 V (OPA2192 / OPA192 supply limit).

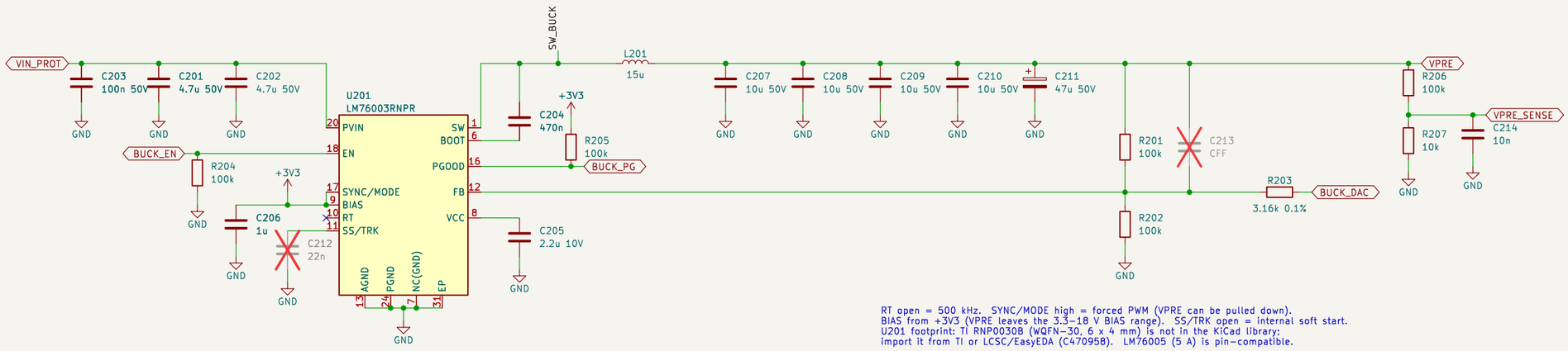
e-psu
embedded-pro
Sheet: /Power input & housekeeping rails/
File: power_input.kicad_sch

Title: Power input & housekeeping rails

Size: A3	Date: 2026-11-03	**Rev: A1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/8

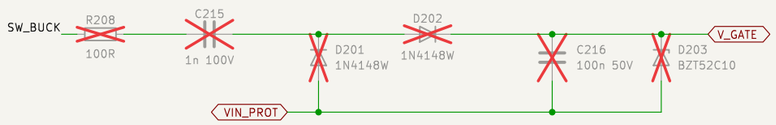
LM76003 tracking pre-regulator: VPRE = VOUT + 1.5 V (2 V … 27.5 V)

VPRE = VFB (1 + R201/R202) + (VFB − V_BUCK_DAC) R201/R203. VFB = 1.006 V
BUCK_DAC 0.2 V –> 27.5 V; 1.0 V –> 2.0 V; -25 mV per G474 DAC step.
MCU in reset: PA4 is high-impedance –> no injection –> VPRE = 2.0 V (safe default).
Firmware rule: VPRE = measured VOUT + 1.5 V; raise VPRE before VOUT, lower VOUT before VPRE.



RT open = 500 kHz. SYNC/MODE high = forced PWM (VPRE can be pulled down).
BIAS from +3V3 (VPRE leaves the 3.3–18 V BIAS range). SS/TRK open = internal soft start.
U201 footprint: TI RNP0030B (WQFN–30, 6 x 4 mm) is not in the KiCad library;
import it from TI or LCSC/EasyEDA (C470958). LM76005 (5 A) is pin-compatible.

Optional gate-drive charge pump V_GATE = VIN + 10 V (not fitted)



Fit to raise the maximum output from ~24 V to ~27 V:
fit R208, C215, D201–D203, C216 and R315 (linear sheet),
remove R314.

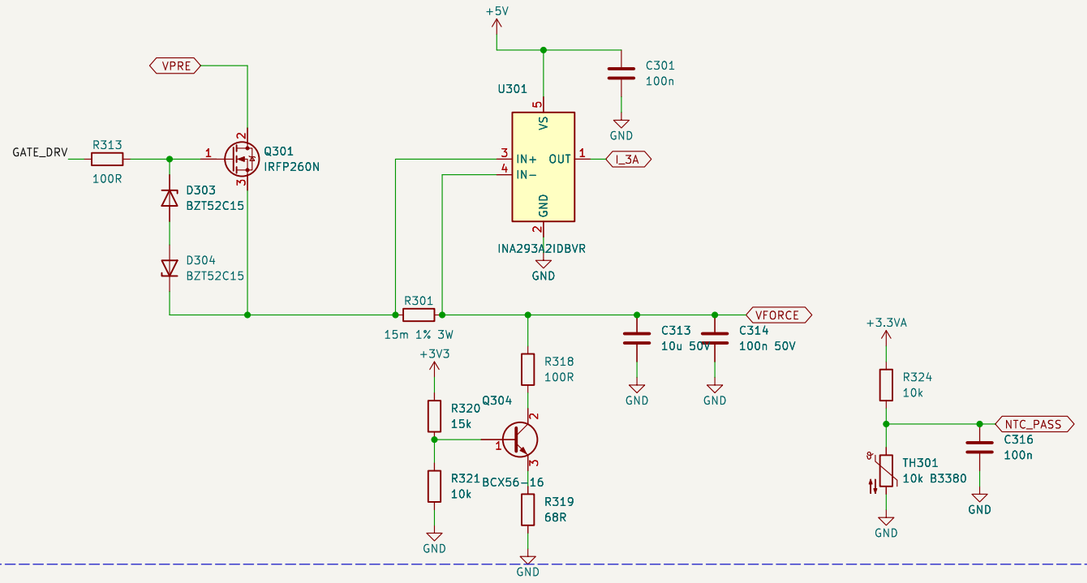
e-psu
embedded-pro
Sheet: /Tracking buck pre-regulator/
File: buck.kicad_sch

Title: Tracking buck pre-regulator

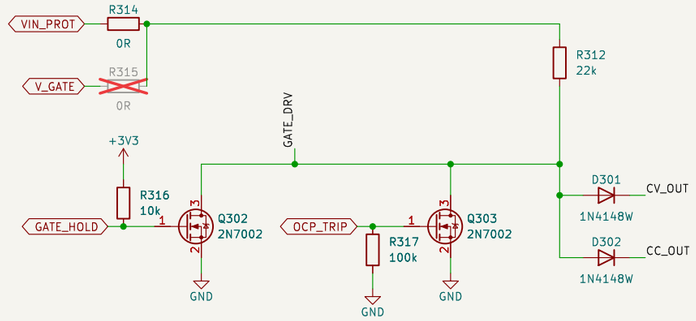
Size: A3	Date: 2026-11-03	**Rev: A1**
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Pass element, CC shunt and force node

I_3A = 15 mOhm x 50 = 0.75 V/A (3.33 A full scale). D303/D304: |VGS| <= 15.7 V.
Q304: 10 mA bleeder at the force node keeps the loop in control at no load
(it sits before the ranging shunts, so it is never measured).

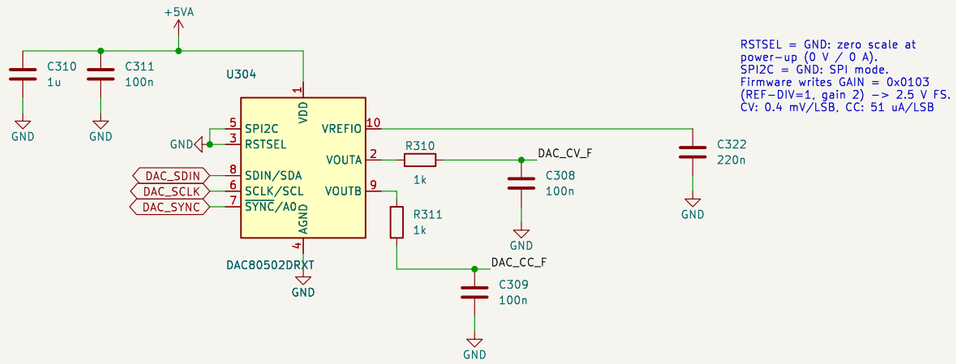


Gate drive and hardware clamps



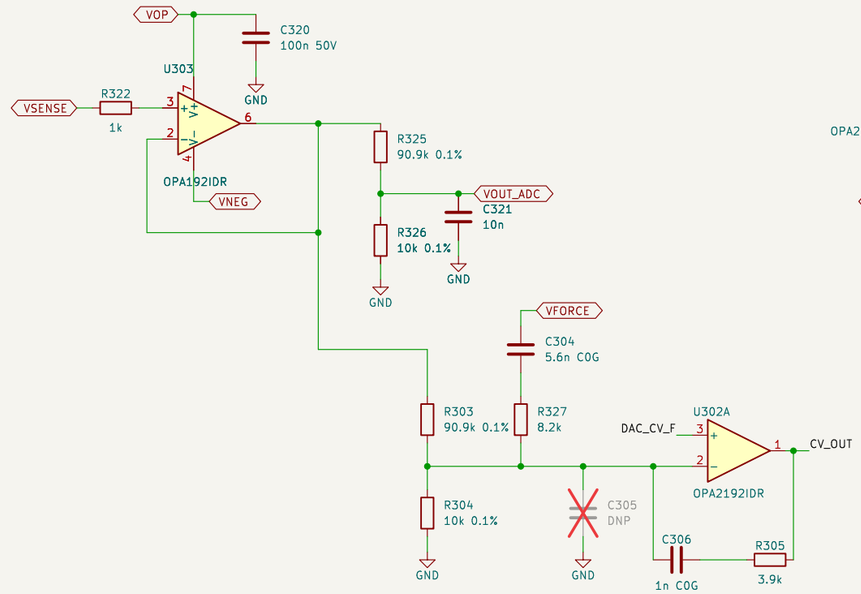
CV / CC amps only pull the gate down (D301, D302); R312 pulls it up. Q302 holds the gate low until
firmware releases GATE_HOLD (pull-up = output off at reset). Q303: comparator trip (TIM20 break, us).

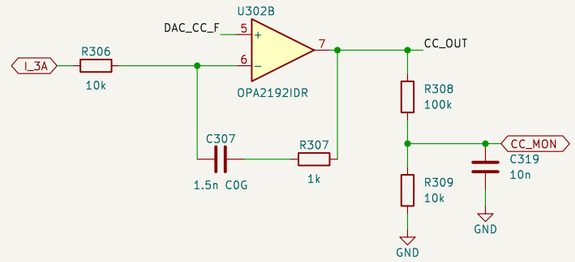
Setpoint DAC: CV and CC references (0 ... 2.5 V)



RSTSEL = GND: zero scale at
power-up (0 V / 0 A).
SPI2C = GND: SPI mode.
Firmware writes GAIN = 0x0103
(REF-DIV=1, gain 2) -> 2.5 V FS.
CV: 0.4 mV/LSB, CC: 51 uA/LSB

Sense buffer (OPA192), CV / CC error amplifiers (OPA2192)





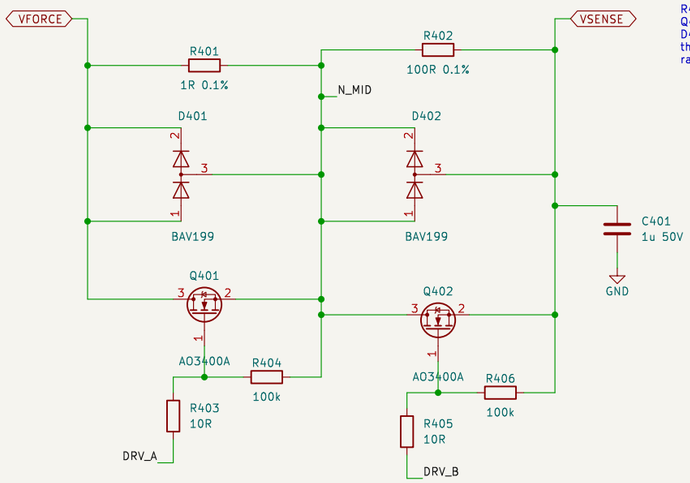
FBV = VS_BUF / 10.09 (DC, sense node): C304 + R327 feed the force node in at AC, so the loop
stays stable in every shunt range. FBI = I_3A (0.75 V/A). Compensation verified in ngspice
(sim/REPORT.md): CV PM >= 35 deg, CC PM >= 91 deg over load, range, C_load 0-1000 uF.

e-psu
embedded-pro
Sheet: /Linear CV/CC stage & setpoint DAC/
File: linear.kicad_sch

Title: Linear CV/CC stage & setpoint DAC

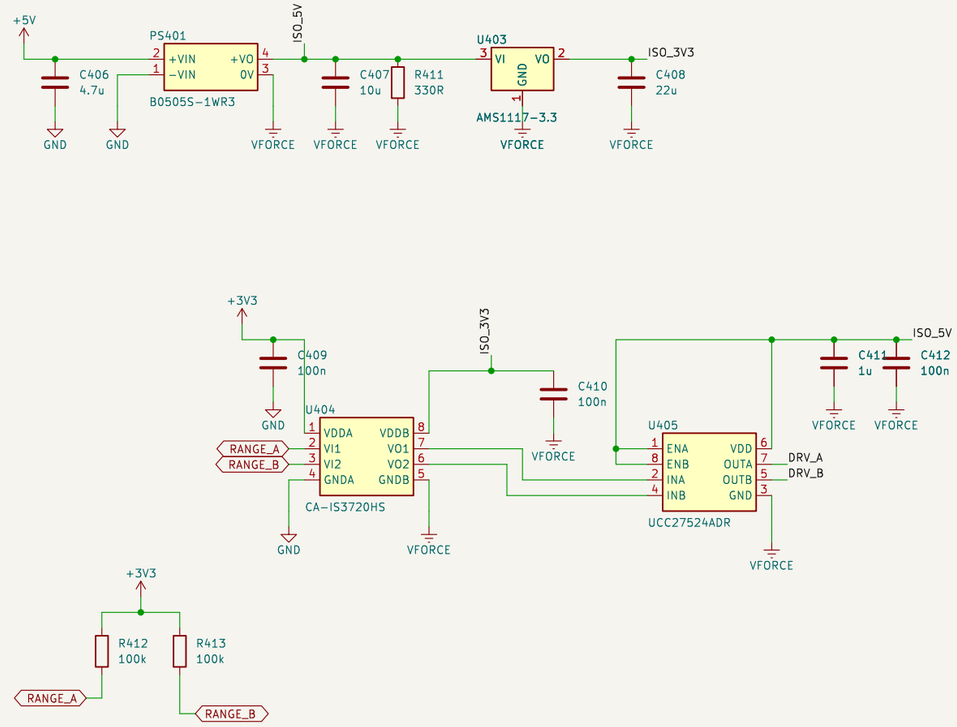
Size: A3	Date: 2026-11-03	**Rev: A1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/8

Shunt chain: force node -> 1 R -> 100 R -> sense node, with bypass FETs



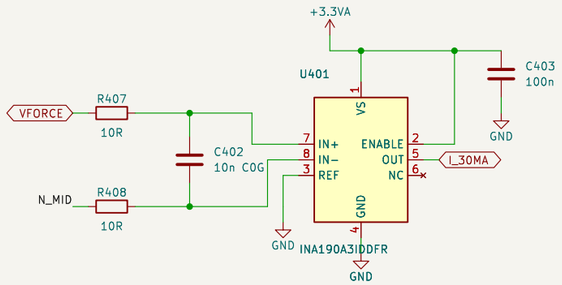
R401 + R402: 30 mA and 300 uA ranges.
Q401 / Q402 short them for the 3 A range.
D401 / D402 (BAV199, 5 nA leakage) cap
the burden at ~0.6 V per shunt while
ranging up. C401: <= 1 uF after the shunts.

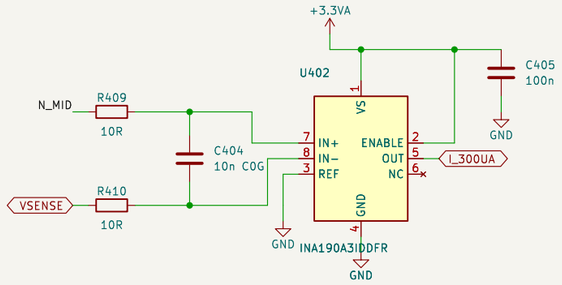
Isolated gate-drive island (reference = VFORCE)



The island floats at the force node, so the bypass FETs get VGS = 5 V at any output voltage.
CA-IS3720HS (suffix H): outputs default HIGH while the MCU does not drive RANGE_A/B (reset)
-> both bypass FETs on -> highest range (lowest burden). R412/R413 make that explicit.
R411: B0505S needs >= 20 mA (10 %) load; the island draws ~8 mA, R411 adds 15 mA.
Keep the island compact; no copper of other nets under PS401 / U404 (isolation).
Range bypass: RANGE_A -> Q401 (1 R), RANGE_B -> Q402 (100 R).

Range sense amplifiers (INA190A3, gain 100, 30 mV full scale)





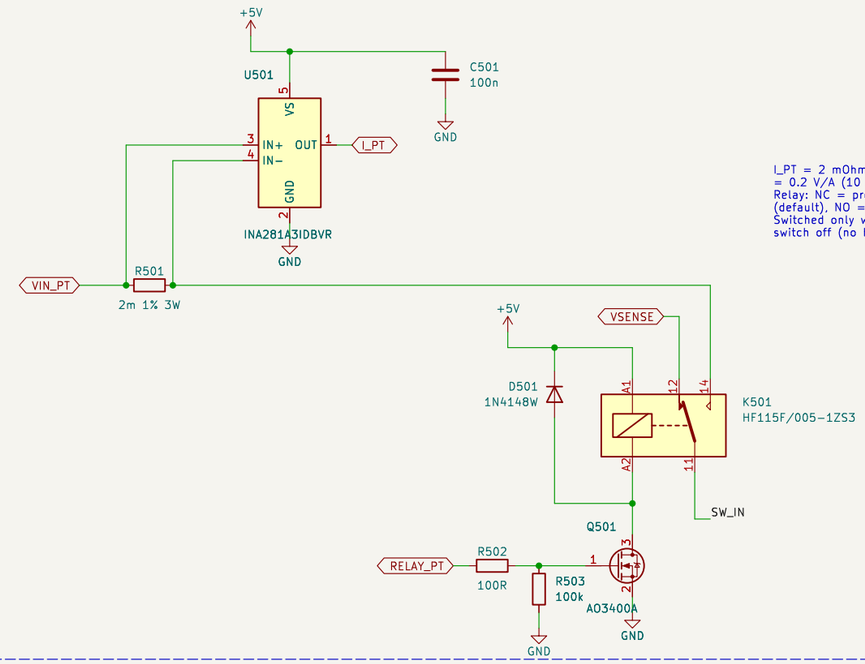
I_30MA: 30 mA -> 3.0 V (8 uA/LSB @ 12 bit). I_300UA: 300 uA -> 3.0 V (80 nA/LSB).
Range-up in hardware: COMP1 (I_300UA) / COMP2 (I_30MA) trip the TIM1 / TIM8 break
-> RANGE_B / RANGE_A latch high within ~1 us; firmware ranges down with hysteresis.

e-psu
embedded-pro
Sheet: /Current measurement & auto-ranging/
File: ranging.kicad_sch

Title: Current measurement & auto-ranging

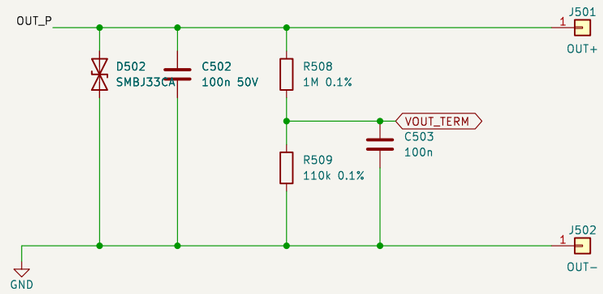
Size: A3	Date: 2026-11-03	**Rev: A1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/8

10 A pass-through measurement and mode relay



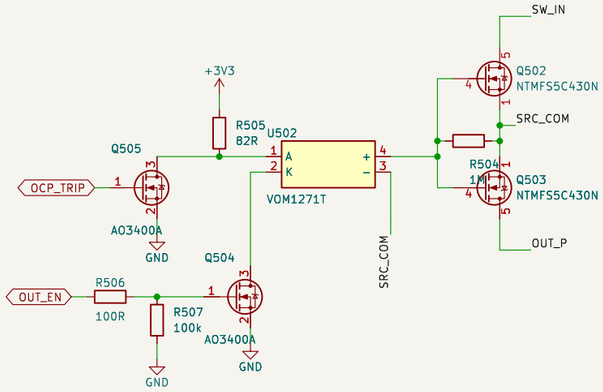
I_PT = 2 mOhm x 100
= 0.2 V/A (10 A -> 2.0 V).
Relay: NC = precision path
(default), NO = pass-through.
Switched only with the output
switch off (no hot switching).

Output terminals and terminal voltage sense



VOUT_TERM = OUT_P / 10.09 at the terminals: firmware trims the CV setpoint for the relay + switch drop (slow outer loop) and subtracts the divider current (VOUT / 1.11 MOhm) from I.

Output switch: back-to-back N-FETs, photovoltaic gate driver



Works at any output voltage (0 ... 30 V).
LED 20 mA -> ~30 uA gate current:
turn-on ~3 ms, turn-off ~0.1 ms.
OCP_TRIP shorts the LED (Q505).
Q502 / Q503 share the source node:
both body diodes block when off.

e-psu
embedded-pro
Sheet: /Pass-through & output switch/
File: output.kicad_sch

Title: Pass-through & output switch

Size: A3	Date: 2026-11-03	**Rev: A1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/8

STM32G474RET6

+3V3

C601 100n
C602 100n
C603 100n
C604 100n
C605 4.7u
C610 100n

GND

One 100 nF per VDD pin (1, 16, 32, 48, 64) close to the pin; 1 uF + 10 nF on VDDA and VREF+.

+3.3VA

C606 1u
C607 10n
C608 1u
C609 100n

GND

Y601
8MHz
OSC_IN
OSC_OUT
C611 27p C0G
C612 27p C0G

GND

NRST
C613 100n

GND

BOOT0
R601 10k

GND

+3V3

J601
SWDIO
SWCLK
SWO
NRST
SWD

GND

TP601 UART_TX
TP602 UART_RX
TP603 GND

U601
+3.3VA
+3V3 +3.3VA
VBAT VDD VDD VDD VDD VDDA
1 16 32 48 64 29

VREF+	28	PA0	12
NRST	7	PA1	13
OSC_IN	5	PA2	14
OSC_OUT	6	PA3	17
PD2	55	PA4	18
NTC_PASS	8	PA5	19
I_3A	9	PA6	20
PC2	10	PA7	21
NTC_BOARD	11	PA8	42
RELAY_PT	22	PA9	43
BUCK_PG	23	PA10	44
RANGE_A	38	PA11	45
BUCK_EN	39	PA12	46
GATE_HOLD	40	PA13	49
OUT_EN	41	PA14	50
DAC_SCLK	52	PA15	51
DAC_SDIN	53	PB0	24
BTN	54	PB1	25
PC13	2	PB2	26
PC14	3	PB3	56
PC15	4	PB4	57
		PB5	58
		PB6	59
		PB7	60
		PB8	61
		PB9	62
		PB10	30
		PB11	33
		PB12	34
		PB13	35
		PB14	36
		PB15	37
VSS	15		
VSSA	27		

PA0 I_300UA
PA1 UART_TX
PA2 UART_RX
PA3 BUCK_DAC
PA7 I_30MA
PA8 RANGE_B
PA10 USB_CONNECT
PA11 USB_DM
PA12 USB_DP
PA13 SWDIO
PA14 SWCLK
PA15 DAC_SYNC
PB0 I_PT
PB1 VPRE_SENSE
PB2 OCP_TRIP
PB3 SWO
PB5 LED_CV
PB6 LED_CC
PB7 LED_OUT
PB8 BOOT0
PB9 LED_PT
PB11 VOUT_ADC
PB12 VIN_SENSE
PB14 CC_MON
PB15 VOUT_TERM

STM32G474RET6
GND

UART log: USART2 (PA2 TX, PA3 RX), 3.3 V. SWD: Cortex 10-pin, SWO on PB3.

User interface

+3V3
R602 10k
BTN
SW601 OUTPUT
C614 100n
GND

LED_CV	R603 1k	D601	green CV
LED_CC	R604 1k	D602	red CC
LED_OUT	R605 1k	D603	yellow OUT
LED_PT	R606 470R	D604	blue PASS

GND

Board temperature and mounting

+3.3VA
R607 10k
NTC_BOARD
C615 100n
TH601 10k B3380
GND

H601 M3
H602 M3
H603 M3
H604 M3

Pin plan (STM32G474RE, LQFP-64):
COMP1+ PA1 I_300UA -> TIM1 BRK -> PA8 RANGE_B
COMP2+ PA7 I_30MA -> TIM8 BRK -> PC6 RANGE_A
COMP3+ PC1 I_3A, COMP4+ PB0 I_PT, COMP6+ PB11 VOUT_ADC
-> TIM20 BRK -> PB2 OCP_TRIP (idle state high)
DAC1_OUT1 PA4 BUCK_DAC; SPI3 PC10/PC12/PA15 -> DAC80502
ADC: PC0 PC3 PB1 PB12 PB14 PB15 (temps, VPRE, VIN, CC_MON, VOUT_TERM)

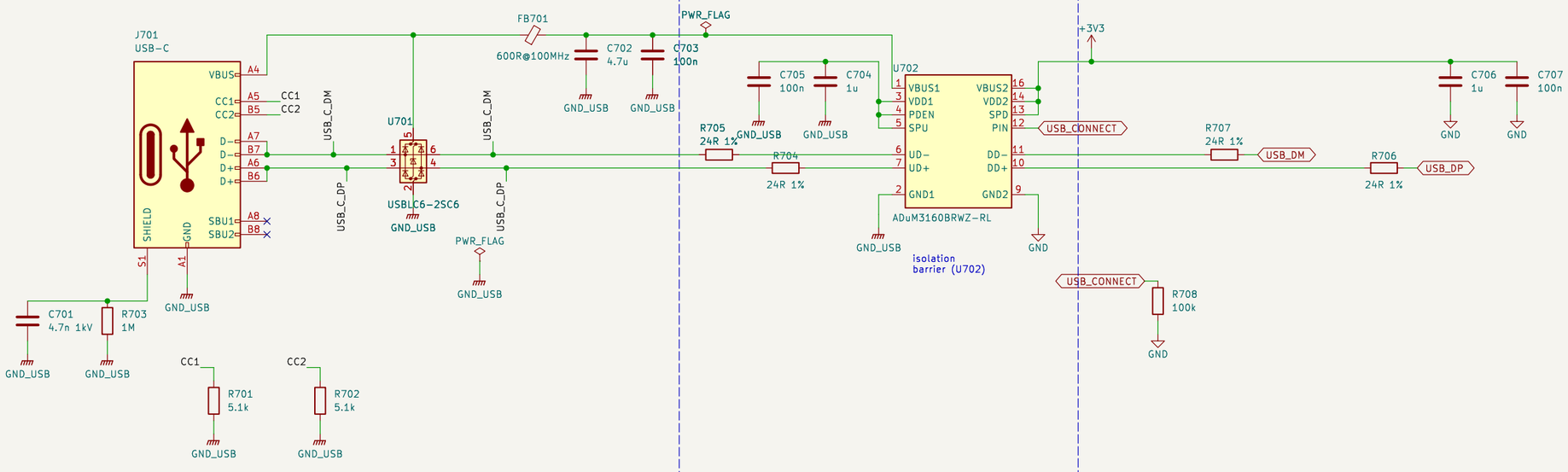
e-psu
embedded-pro
Sheet: /MCU, debug & user interface/
File: mcu.kicad_sch

Title: MCU, debug & user interface

Size: A3	Date: 2026-11-03	**Rev: A1**
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PC side (GND_USB, powered from the USB cable)

e-psu side (GND, +3V3)



Full speed: SPU = SPD = high. PDEN = VDD1 (normal operation). 24 R +/-1 % on all four data lines.
USBLC6 pins 1-6 / 3-4 are flow-through pairs: route D-/D+ across the pads.

USB_CONNECT (PA10) enables the PC-side pull-up only when the USB stack is ready
(re-enumerate by pulling it low for ~100 ms). VBUS2 = VDD2 = +3V3 (regulator bypassed).
Layout: >= 8 mm creepage under U702, no copper on any layer crossing it; 90 Ohm D+/D- pairs.

e-psu		
embedded-pro		
Sheet: /Isolated USB/		
File: usb.kicad_sch		
Title: Isolated USB		
Size: A3	Date: 2026-11-03	**Rev: A1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/8